Ankit Kumar Jha

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Summary

AI & ML engineer with a B.Tech in Artificial Intelligence & Machine Learning who designed FinSight, a dual-retrieval (dense + BM25) RAG pipeline with cross-encoder reranking for financial document Q&A. Built LLM evaluation frameworks at Ethara AI and shipped 3+ end-to-end ML pipelines as a freelancer. Seeking an AI/LLM Engineer, Data Analyst and Data Scientist role to build and evaluate applied generative AI systems.

Education

Guru Gobind Singh Indraprastha University — New Delhi, India
Bachelor of Technology in Artificial Intelligence & Machine Learning; CGPA: 8.32 — *Oct 2022 – Jun 2026*

- Coursework: Machine Learning, Deep Learning, NLP, Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Optimization Theory

Technical Skills

LLM/GenAI: Generative AI, RAG (hybrid dense+BM25 retrieval, cross-encoder reranking, table-aware chunking), Hugging Face Transformers, LangChain, LangGraph, Prompt Engineering, Fine-tuning (LoRA/PEFT), GPT-4, LLaMA
Languages: Python, SQL, Java, C++
Software Engineering: OOPs, LLD, SDLC
Core ML: PyTorch, TensorFlow, Keras, Scikit-learn, Feature Engineering, Time-Series Forecasting (LSTM – StockAI), Bayesian Inference, Convex Optimization, Information Theory
Deployment: Docker, FastAPI, Streamlit, AWS (EC2, S3), GCP (Vertex AI), Git, CI/CD, REST APIs
Databases: MySQL, PostgreSQL, MongoDB, ChromaDB (dense vector search), rank-bm25 (sparse search, FinSight)
Data Analysis: Excel, Power BI, Pandas, NumPy

Projects

FinSight | *Python, RAG, Hugging Face, FastAPI, Streamlit, Docker* — 2026

- Built a dual-retrieval RAG pipeline (dense + BM25) with cross-encoder reranking for financial document Q&A, with table-aware chunking to preserve numeric context; improved Recall@5 from 68% to 81%.
- Achieved 87% answer faithfulness on a 40-question evaluation set with source citations; containerized and deployed to Hugging Face Spaces via Docker.

Conversational RAG Assistant | *LangChain, Mistral, ChromaDB, Streamlit* — 2026

- Built a multi-turn conversational RAG chatbot with memory-aware query rewriting, extending FinSight's single-shot retrieval to handle follow-up questions and context carryover.
- Implemented cosine-similarity vector search (ChromaDB) over 10K+ documents; measured hallucination rate at 14% vs. a vanilla-LLM baseline of 35% (50-question test set).

Churn Modelling | *Python, XGBoost, LightGBM, Optuna, FastAPI, Docker, MLflow* — Ongoing

- Developing an end-to-end Customer Churn Prediction system on 10,000 bank customers by engineering a multi-sheet relational pipeline (Demographic, Location, Account) with scikit-learn ColumnTransformer, benchmarking 8 ML algorithms (LR, RF, XGBoost, LightGBM, SVM, etc.) via Optuna-powered hyperparameter tuning, and deploying a real-time inference API (FastAPI) containerized with Docker.
- Enhancing model robustness on a 36% missing-value feature (Balance) through controlled experimentation (drop vs. median imputation vs. missing indicator), tracking all experiment metrics with MLflow, and structuring the project as a modular, production-grade codebase with automated scripts for data ingestion, preprocessing, training, and model registry.

Experience

LLM Intern — Feb 2026 – Sept 2026
Ethara AI — *Gurugram, Haryana, India*

- Developed Python-based automation scripts for large-scale LLM output evaluation, reducing manual assessment time by 50% and enabling consistent scoring across 1,000+ prompt variations.
- Designed structured evaluation frameworks with custom metrics (accuracy, coherence, hallucination rate), cutting new-model onboarding from 5 days to 2 days.
- Benchmarked 5+ LLMs across production use cases, producing reports that directly informed the team's model-selection decisions.
- Developed and implemented multi-stage testing pipelines to identify and diagnose performance issues in LLM applications, delivering technical reports that guided root cause analysis and resolution for stakeholders.

Freelance Full-Stack ML Engineer — Jan 2025 – Jun 2026
Independent (part-time, alongside full-time coursework) — *Remote*

- Architected and delivered 3+ end-to-end ML pipelines (ingestion → feature engineering → training → deployment) across NLP, classification, and time-series domains, achieving 85%+ accuracy on a customer-churn task, a 12-point lift over baseline (73%).
- Deployed containerized ML services via Docker on AWS EC2 and GCP for 45+ daily active users; built the accompanying ETL layer, and debugged service logs to identify and resolve operational issues.
- Diagnosed and resolved API integration and deployment issues in deployed ML services, improving system stability for users.
- Responded to client inquiries on ML pipeline performance and bugs, communicating solutions via email and documentation.
- Authored technical documentation for deployed ML pipelines, including troubleshooting guides and API usage instructions.
- Implemented advanced feature engineering (polynomial features, PCA, interaction terms) on 10K+ record datasets to improve model performance and reduce training time.

Certifications & Achievements

- **Google Cloud ML Specialization** (Vertex AI); **Deep Learning & NLP Specializations** (Coursera)
- **Active GitHub contributor** – ML projects spanning RAG pipelines, evaluation frameworks, forecasting models